



What Data Are Normal?

Only certain data sets have a normal distribution. Looking at the graph of the data can help to determine if the data set is normally distributed.

Determining Whether a Data Set Is Normal

You can determine whether a data set has a normal, or approximately normal, distribution by creating a display of the data.

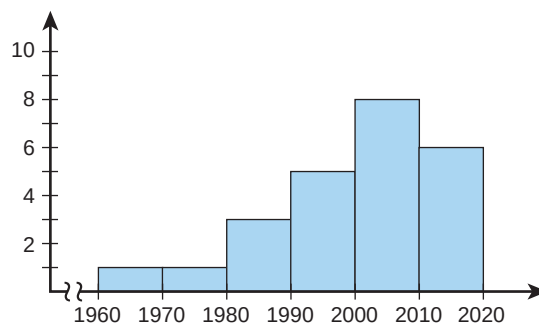
EXAMPLE

Here are the dates of 24 pennies found in a piggy bank. Explain whether the data are normally distributed.

1984, 2012, 2013, 1999, 2013, 2010, 2009, 1987, 2002, 1966, 1979, 2009, 2012, 2008, 1992, 1991, 1998, 2003, 2005, 2007, 1989, 1997, 2006, 2014

SOLUTION

The dates are widely spread, so a histogram is an appropriate display.



The graph is skewed left, so the data are not normally distributed. ■

- **Think About It** For a small data set with a small range of values, a line plot or stem plot would be appropriate.

Standard Deviation and Nonnormal Data

The mean of the penny data set is about 2000, and the standard deviation is about 12. If the data set were normally distributed, you could conclude that about 99.7% of the pennies, or nearly all the pennies, have dates between 1964 and 2036. Although it is true that all the dates in the data set are between 1964 and 2036, it is a misleading statement because the greatest date in the list is 2014.

- **Remember** In a normal distribution, the mean, median, and mode are the same.